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JACC STATE-OF-THE-ART REVIEW

Interpreting the Kansas City Cardiomyopathy Questionnaire in Clinical Trials and Clinical Care

JACC State-of-the-Art Review

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ABSTRACT

To improve the patient-centeredness of care, patient-reported outcomes have been increasingly used to quantify patients' symptoms, function, and quality of life. In heart failure, the Kansas City Cardiomyopathy Questionnaire (KCCQ) has been qualified by the U.S. Food and Drug Administration as a Clinical Outcome Assessment and recommended as a performance measure for quantifying the quality of care. By systematically asking the same questions reproducibly over time, the KCCQ can validly and sensitively capture the impact of heart failure on patients' lives and is strongly associated with clinical events over time. This review describes how to interpret the KCCQ, how it should be analyzed in clinical trials to maximize the interpretability of results, and how it can be used in clinical practice and population health. By providing a deeper understanding of the KCCQ, it is hoped that its use can further improve the patient-centeredness of heart failure care. (J Am Coll Cardiol 2020;76:2379–90) © 2020 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

The primary goals of treating heart failure are to minimize disease progression (death, hospitalization) and to optimize patients' health status—their symptoms, function, and quality of life. As such, clinical trials focus on these outcomes to establish the evidence base with which to guide patient care. Whereas death and hospitalization are easy to measure, health status can be more difficult to capture in a reliable and valid manner. In the past, this has been most often assessed in trials using the New York Heart Association (NYHA) functional class (1). Whereas NYHA functional class information is simple to collect, it has been shown that health status captured directly from patients is not only a more reliable, sensitive, and valid assessment of the patient's symptoms and function, but it is also more predictive of heart failure hospitalization and death (2–4).

Recently, the U.S. Food and Drug Administration (FDA) has endorsed the view that a treatment is potentially approvable based on improving health status alone (5). Moreover, both the cardiovascular drug and device divisions have qualified the Kansas City Cardiomyopathy Questionnaire (KCCQ) as a Clinical Outcome Assessment (6,7), meaning that approval and labeling claims can be obtained by demonstrating clinically important improvements on the KCCQ. Thus, tools like the KCCQ are likely to become more commonly used in clinical trials. Whereas interpreting clinical events such as death

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ABBREVIATIONS AND ACRONYMS

CI = confidence interval

FDA = U.S. Food and Drug Administration

HFpEF = heart failure with preserved ejection fraction

HFrEF = heart failure with reduced ejection fraction

KCCQ = Kansas City Cardiomyopathy Questionnaire

MCID = minimal clinically important difference

METs = metabolic equivalents

NNT = number-needed-to-treat

NYHA = New York Heart Association

and hospitalization is relatively straightforward, the interpretation of health status outcomes is less obvious. This is, in part, because they are continuous rather than dichotomous measures and, in part, because the scales are not intuitively familiar. Therefore, a solid framework for interpreting health status outcomes, such as the KCCQ, in clinical trials is needed so that the results can be meaningfully understood and communicated.

Beyond their role as outcomes in clinical trials, patient-reported health status outcomes also have an important potential role in clinical care (8). As a means of reproducibly asking relevant questions about the frequency and severity of symptoms, the physical and social limitations associated with those symptoms, and patients' perceptions of the impact of their symptoms and function on their quality of life, the KCCQ performs the role of a "standardized history," a fundamental component of each clinical encounter. Not only is this type of standard assessment helpful for tracking patients over time to identify changes in patients' health status warranting clinical action, it can also be particularly useful to health care systems where patients are seen by multiple providers to provide a more consistent insight into the patient's well-being. Moreover, by explicitly quantifying the health status of all patients with heart failure in a health system, system-level interventions can be implemented to support population health. In the age of the coronavirus disease-2019 pandemic, where tele-health became rapidly adopted worldwide, monitoring patients' status over time in a reliable and efficient manner is more important than ever, and a better understanding of the meaning of KCCQ scores offers the opportunity to better manage patients with chronic heart failure.

To address these challenges in the understanding of the KCCQ as an outcome in clinical trials and as a tool for patient care, we sought to provide a detailed review of the instrument's interpretability. Because these different applications have different, but overlapping, contexts, the perspectives of both are presented. These conceptual frameworks are also potentially applicable to other patient-reported outcomes. Collectively, this information is intended to provide a more detailed insight into how best to interpret patients' health status and lay the foundation for enabling patients' experiences to become a more integral part of evaluating novel therapies and improving clinical care.

HIGHLIGHTS

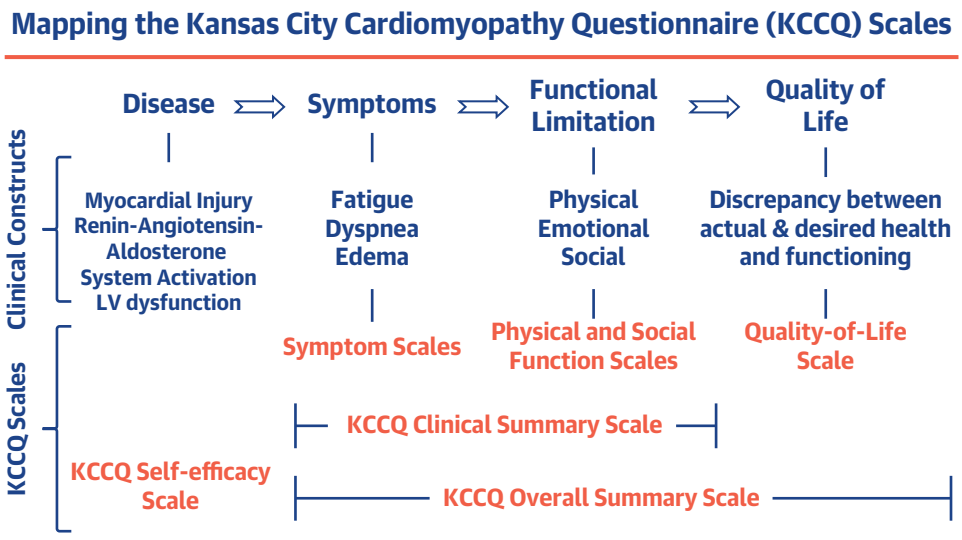
- The KCCQ measures symptoms, physical and social limitations, and quality of life in patients with heart failure.
- In clinical trials, mean differences in KCCQ scores are more difficult to interpret than the proportion of patients experiencing clinically important changes.
- Broader understanding of the meaning of KCCQ scores might encourage greater use of the tool to support patient-centered care.
- Research is needed on how best to integrate the KCCQ into practice and assess its impact on patient care.

OVERVIEW OF THE KCCQ

Developed in 1996 and published in 2000 (9), the KCCQ was designed with input from patients and clinicians to capture those domains of how heart failure affects patients' lives. The KCCQ has a 2-week recall period (given the day-to-day variability in heart failure symptoms) and includes 23 items that map to 7 domains (**Central Illustration**): symptom frequency; symptom burden; symptom stability; physical limitations; social limitations; quality of life; and self-efficacy (the patient's understanding of how to manage their heart failure). The symptom frequency and symptom burden domains are merged into a total symptom score, which can be combined with the physical limitation domain to create a clinical summary score that mirrors the key concepts of the NYHA functional class. The symptom, physical limitations, social limitations, and quality of life domains can also be combined to create an overall summary score, which has been the primary health status outcome for the KCCQ in most trials. The KCCQ was designed to capture each of the domains in a simple manner that was comparably assessed in men and women, older and younger patients, and across the range of patients' socioeconomic status. To facilitate interpretation, all scores are represented on a 0-to-100-point scale, where lower scores represent more severe symptoms and/or limitations and scores of 100 indicate no symptoms, no limitations, and excellent quality of life.

To make the KCCQ more feasible to implement in routine clinical care, it was reduced from its original 23 items (KCCQ-23) into a 12-item instrument

CENTRAL ILLUSTRATION **Conceptual Mapping of the Kansas City Cardiomyopathy Questionnaire to Different Manifestations of Heart Failure**



Spertus, J.A. et al. J Am Coll Cardiol. 2020;76(20):2379–90.

Whereas there is an underlying disease process causing the heart failure syndrome, patients are aware of the symptoms that they experience and how those symptoms affect their ability to function and quality of life. Different scales and summary scores map to different manifestations of heart failure, as demonstrated in the figure. KCCQ = Kansas City Cardiomyopathy Questionnaire; LV = left ventricular.

(KCCQ-12), which includes the symptom frequency, physical limitations, social limitations, and quality of life domains and can also generate the clinical and overall summary scores with excellent concordance to the respective scores of the full instrument (10). Choosing between the 23-item and 12-item versions of the KCCQ depends on the intended use of the instrument. Should the symptom stability or self-efficacy scales be desired, then the 23-item version should be selected, as these are not in the abbreviated questionnaire. In addition, both the drug and device divisions of the FDA have qualified the longer, but not yet the shorter, version of the KCCQ, despite having been provided with identical psychometric data for both forms. Accordingly, clinical trials seeking FDA approval for new treatments may choose the longer version of the KCCQ. In most other circumstances, including routine clinical care, the KCCQ-12 would likely be preferred due to the lower response burden on patients.

Both versions of the KCCQ have undergone linguistic and cultural translation into >100 languages and/or countries (11). For simplicity, this review focuses on the overall summary score, but the concepts presented are readily applied to all of the individual scales. The psychometric properties of the KCCQ (its validity, reliability, and responsiveness) have been demonstrated in numerous etiologies of heart failure, including heart failure with preserved ejection fraction (HFpEF) and heart failure with reduced ejection fraction (HFrEF) and valvular disease, and are not repeated here (10,12–14).

INTERPRETING THE KCCQ AT A POINT IN TIME

To better understand what specific scores at a single point in time mean, scale scores can be interpreted within a clinical context or by their association with prognosis.

CLINICAL CORRELATES OF KCCQ SCORES. As described, all KCCQ scores are scaled from 0 to 100 and frequently summarized in 25-point ranges, where scores represent health status as follows: 0 to 24: very poor to poor; 25 to 49: poor to fair; 50 to 74: fair to good; and 75 to 100: good to excellent. Because the most common means of quantifying health status in clinical practice and trials is the NYHA functional class, it is valuable to appreciate what KCCQ scores correlate to which NYHA functional class. There is no

exact mapping for such a conversion, primarily because of the well-documented inaccuracies of quantifying the NYHA functional class (1,15). In fact, a recent study highlighted the marked variability in the assignment of NYHA functional class to patients with similar KCCQ scores in clinical trials where the NYHA functional class was used as a criterion for patient eligibility (16). Nevertheless, to provide a context for interpreting the ranges of KCCQ overall summary scores in the context of the NYHA functional class, about 85% of patients with scores of 0 to 24 are NYHA functional class III/IV; 60% of patients with scores of 25 to 49 are NYHA functional class III; one-half of patients with scores of 50 to 75 are NYHA functional class III and one-half are NYHA functional class II; and of those with scores over 75, over 80% are NYHA functional class I or II.

PROGNOSTIC SIGNIFICANCE OF CROSS-SECTIONAL KCCQ SCORES. The association of KCCQ scores with prognosis has been demonstrated at the time of discharge from hospitalization for acutely decompensated heart failure (13), 1 week after hospital discharge (17), in the outpatient setting (3,12,18), and before and after percutaneous valvular intervention (19). In all these studies, there was an independent association among KCCQ scores and death, heart failure hospitalization, or a composite of either event, even after adjusting for numerous other clinical characteristics. The independent association of KCCQ scores with clinical events is because the KCCQ captures an important domain that is not represented by any other clinical markers of disease severity, such as ejection fraction or B-type natriuretic peptide (20). When compared with patients having scores of 75 to 100, those with scores of 50 to 74, 25 to 49, and 0 to 24 have a relative risk of death or hospitalization of 1.5, 2, and 3, respectively; these are ratios that are relatively consistent in both outpatient and inpatient settings (3,13). Given the independence of these associations from patient factors, when evaluating 2 patients with an otherwise similar estimated risk of clinical events, a patient with a KCCQ score <25 is 3× as likely to experience death or hospitalization as a patient with similar clinical characteristics whose KCCQ score is >75.

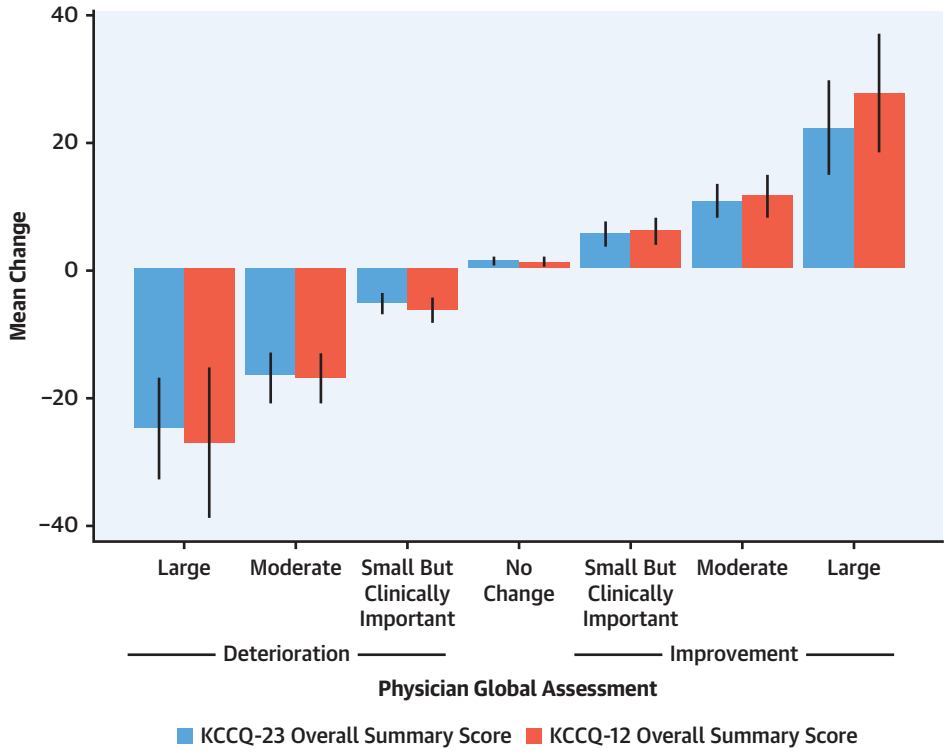
INTERPRETING CHANGES IN KCCQ SCORES

A good heuristic for interpreting changes in KCCQ scores is to establish a threshold of change that is clinically important. As will be developed further, a change of 5 points is considered to be a small but clinically important change, whereas changes of 10 and 20 points are considered moderate-to-large and large-to-very large clinical changes. Several streams of evidence support these interpretations.

AVERAGE CHANGES IN KCCQ SCORES FOR POPULATIONS OF PATIENTS. The classic approach to estimating the clinical importance of changes in scores is to define groups of patients who have small but clinically important (the minimal clinically important difference [MCID]), moderate, and large changes in their health status (as estimated by clinicians or patients) and to compare the mean differences in scores among these groups. Following this model, a 14-center study was explicitly designed to estimate these thresholds of clinically important changes (4). In this study, clinicians' assessments of clinical change revealed great symmetry and proportionality for both improvements and deteriorations in patients' health status and suggested the thresholds of 5, 10, and 20 points represent small, moderate-to-large, and large-to-very large clinical changes. Importantly, these thresholds applied equally well for both improvements and deteriorations and, as shown in **Figure 1**, are exceedingly similar for both the KCCQ-23 and KCCQ-12 (2). Moreover, there is great similarity for the magnitudes of the prognostic association between changes in KCCQ scores for both improvements and deterioration (2,12). Notably patients' self-classifications of clinically important improvements were associated with even smaller mean changes in KCCQ scores (21). A more recent analysis from a clinical trial also suggested smaller mean changes are clinically important (22). In that study, Butler et al. (22) found that patients reporting small improvements had 3.6 points (95% confidence interval [CI]: 1.0 to 6.2 points) and 4.3 points (95% CI: 0.2 to 0.4 points) greater KCCQ overall summary scores than did patients reporting no change in their heart failure status at 4 and 24 weeks after randomization in a clinical trial of iron supplementation in HFrEF. The corresponding differences between those reporting a slight worsening in their condition, as compared with those who did not change, were −0.4 points (95% CI: −8.6 to 7.7 points) and −5.0 points (95% CI: −15.5 to 5.6 points), respectively.

FACE VALIDITY ESTIMATES OF CHANGE AT AN INDIVIDUAL PATIENT-LEVEL. Additional insight into clinically important changes is gained by understanding the impact of changing 1 category within an item on the score of both that domain and the relevant summary scores. **Figure 2** shows the impact shifting responses on each domain and summary score for both the KCCQ-23 (**Figure 2A**) and the KCCQ-12 (**Figure 2B**).

FIGURE 1 Mean Differences in KCCQ Scores Among Patients With Different Magnitudes of Clinical Change



The mean differences in KCCQ overall summary scores for the 23- and 12-item versions of the Kansas City Cardiomyopathy Questionnaire (KCCQ) are shown for groups of patients assessed by their physicians as having experienced different magnitudes of clinical change.

To more simply understand how to interpret these figures, one can consider the physical limitation scale of the KCCQ-23. The 6-item physical limitation scale of the KCCQ-23 consists of 2 items representing low (<4 metabolic equivalents [METs]), moderate (5 to 7 METs), and high (>6 METs) levels of physical activity and uses a 5-point ordinal scale to grade the level of difficulty, ranging from extremely limited to not limited at all, associated with each activity. To support the disease-specificity of the KCCQ, there is also an option for the patient to state that they did not do the activity or were limited by something else (e.g., arthritis), and these responses are treated as missing variables. Each "shift" in a category of response will change the physical limitation score by 4.2 points and it is not possible to change by <4.2 points. For example, if a patient improved from being moderately to slightly limited in walking a block on level ground, and responses to all other items in the KCCQ physical limitation scale did not change between assessments, then their physical limitation score would increase by 4.2 points. If, however, they were no longer limited at all in walking a block on level ground (a shift of 2 steps in the item response) and no other responses changed, then their KCCQ physical limitation score would increase by 8.3 points. The interpretation of whether a single shift in response or 2 shifts in responses is "clinically important" is somewhat in the eyes of the beholder, although the likelihood of measurement error is less for the latter. Thus, a change of 5 points on the KCCQ physical limitation score would require a shift, in the same direction of at least 2 categories: either 2 categories within a single item or 1 category in 2 different items. For the KCCQ-12, which selected 1 activity within each activity level, each shift in response is associated with an 8.3-point change in score on the physical limitation domain score (Figure 2B).

For the KCCQ-23 overall summary score, the impact of a change in a response item on each question has a much smaller impact, ranging from 0.53 points for the symptom frequency items that have 7 options to 2.1 points for the quality-of-life items. For the KCCQ-12, the impact of each shift in response

ranges from 1 point for the symptom frequency estimates of shortness of breath and fatigue, to 3.1 points for the quality-of-life items. While multiple combinations of changes in responses can lead to changes in scores of 5 points or more, it requires anywhere from 3 to 5 "shifts" on either the KCCQ-23 or the KCCQ-12, which most would consider to be an important change in health status, although patients often consider smaller changes to be clinically important (21). Examining the magnitude of shifts required to move the overall summary score by 10 or 20 points also gives a perspective on how clinically large those changes are.

CLINICAL CORRELATES OF CHANGES IN KCCQ SCORES. Supporting the 5-point thresholds of meaningful clinical change in the KCCQ, numerous studies have demonstrated its association with other clinically important outcomes. The first of these, performed in 2007 by Kosiborod et al. (2), examined the association of changes in KCCQ scores with subsequent cardiovascular mortality and all-cause hospitalization rates. They found that each 5-point improvement in KCCQ scores was associated with an ~10% relative reduction in the risk of adverse clinical events. Conversely, each 5-point decrement in KCCQ scores increased risk by ~10% for both cardiovascular death and the combined endpoint of cardiovascular death and hospitalization. Importantly, the association of change in KCCQ and outcomes was statistically linear, suggesting no variation in the risks by the magnitude of clinical change. These associations have subsequently been confirmed in both HFpEF and HFrEF left ventricular ejection fraction (12).

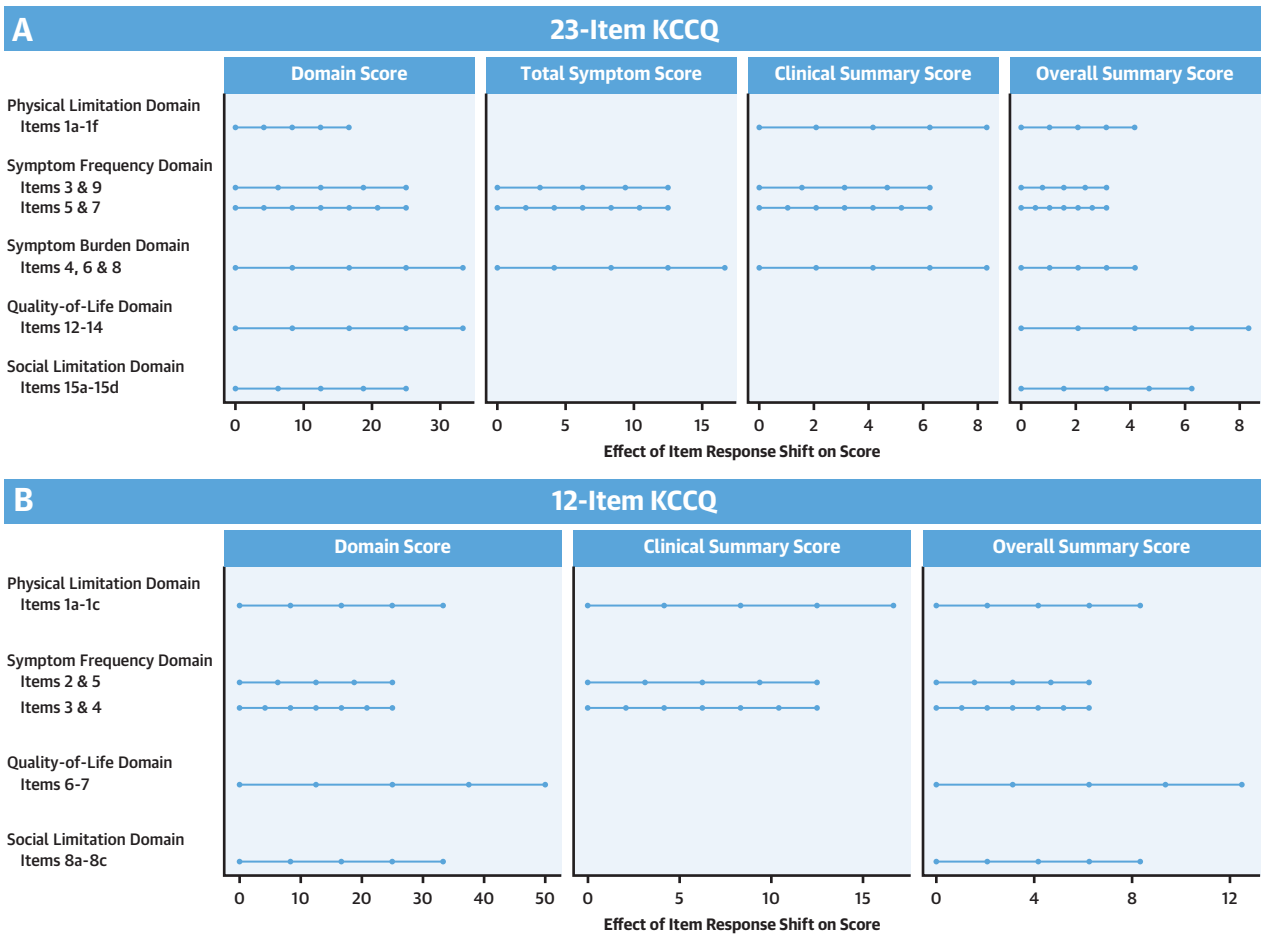
Beyond their prognostic ability, changes in KCCQ scores have been associated with changes in functional capacity, as assessed by both the 6-min walk test and formal cardiopulmonary exercise testing. In the HF-ACTION (Heart Failure: A Controlled Trial Investigating Outcomes of Exercise Training) trial of patients with HFrEF testing the safety and efficacy of structured exercise, Flynn et al. (23) found that a 5-point change in the KCCQ overall summary score was associated with a mean improvement in 6-min walk distance of 112 m (95% CI: 96 to 134 m) and a change of Vo_2 of 2.5 ml/kg/min (95% CI: 2.21 to 2.86 ml/kg/min), although the use of an exercise intervention may have amplified improvement in 6-min walk distances associated with changes in the KCCQ.

Table 1 summarizes these estimates across ranges of change in KCCQ overall summary scores, but a simple approximation would be that each 5-point improvement in KCCQ is associated with a 10% reduction in the risks of mortality and hospitalizations, a 100-m increase in 6-min walk distance, or a 2.5-ml/kg/min increase in peak oxygen consumption. Given that most would consider these to be clinically important relationships, they provide greater confidence in the use of 5-, 10-, and 20-point thresholds of change in the KCCQ for defining those who have experienced small but important, moderate-to-large, and large-to-very large clinical changes.

INTERPRETING THE KCCQ AS AN OUTCOME IN CLINICAL TRIALS

Most clinical trials test the effect of an intervention on health status by comparing the difference in mean scores (or changes in scores) between groups. Examples of studies that have focused on the population-level average differences between groups are SHIFT (Effects of Ivabradine on Cardiovascular Events in Patients With Moderate to Severe Chronic Heart Failure and Left Ventricular Systolic Dysfunction) (mean difference in KCCQ scores of 2.4 points) (24), PARADIGM-HF (Study to Evaluate the Efficacy and Safety of LCZ696 Compared to Enalapril on Morbidity and Mortality of Patients With Chronic Heart Failure) (1.3 points) (25), HF-ACTION (1.9 points) (26), and TOPCAT (Treatment of Preserved Cardiac Function Heart Failure With an Aldosterone Antagonist) (1.5 points) (27). All of these mean differences were statistically significant favoring treatment. Whereas it is tempting to interpret such differences in the context of the MCID, this is not the most accurate or clinically intuitive way to appreciate the benefits of therapy. As discussed, studies have suggested that 5 points represent a small but meaningful change in the KCCQ overall summary score. If this metric is applied to the above-mentioned trials, one would conclude that in all cases the benefit of treatment on health status was statistically significant but clinically inconsequential. However, we assert that this is almost always a misuse of the MCID, arising from attempting to equate between-group mean differences and within-patient change. The only situation where the MCID is applicable in interpreting the mean difference between groups is when there is no confounding (randomization to treatment) and when there is no heterogeneity of treatment effect across patients. We believe that, particularly with patient-centric measures such as health status, heterogeneity is the norm rather than the exception. In such a case, treatment group means represent amalgams of varying outcomes across patients and the difference in means, though broadly describing the shift in distribution of scores, is not applicable to

FIGURE 2 **Impact of "Shifting" 1 Response Item on KCCQ Scores**



Depending on the domain, responses that are 1 category higher or lower will affect the corresponding scores by different amounts. The number of net shifts can be multiplied by the corresponding points to calculate the net difference in scores. The clinical significance of these changes is acquired by examining the items themselves. The effects for the 23-item Kansas City Cardiomyopathy Questionnaire (KCCQ) **(A)** and for the 12-item KCCQ **(B)** are described.

individual patients. This point is represented diagrammatically in **Figure 3**. In **Figure 3A**, the population-mean differences in scores (the average change in scores of the patients who improve [green], stay the same [yellow], or worsen [red]) are not clinically interpretable when trying to apply the results of a trial to clinical practice, as this average effect does not apply to any individual within the population (no patient has a green head, yellow body, and red legs).

One way to address this issue is to expect heterogeneity and to account for it in the analysis. We believe that this is especially critical for patient-reported health status outcomes such as the KCCQ. Frequently, the biggest determinant of improvement in health status is the patient's initial health status, and treatments are often likely most effective among those who have the most to gain. In this light, we recommend that analyses of health status outcomes such as the KCCQ should, as a rule, include a plan to evaluate outcomes and treatment effects as a function of baseline health status (28,29). Accounting for measurable heterogeneity likely only partially moves the between-group comparison of means toward a more patient-specific interpretation; in fact, the majority of sources of heterogeneity are likely unobserved, thus leaving mean differences difficult to interpret.

To provide better clinical interpretability, if a statistically significant difference in mean KCCQ scores is observed, it is strongly encouraged that investigators then describe the distribution of patients experiencing 5-, 10-, or 20-point changes (**Figure 3B**). This enables the proportions of patients treated with one strategy versus another who experienced a

TABLE 1 Clinical Correlates of Changes in KCCQ Scores

	Change in 6-Min Walk Test (m)	Change in Vo_2 (ml/kg/min)	OR for Mortality/Hospitalization (%)	OR for Mortality (%)
Small change (5 points)	112	2.5	11	9
Moderate to large change (10 points)	225	5	23	19
Large to very large change (20 points)	450	10	52	42

KCCQ = Kansas City Cardiomyopathy Questionnaire; OR = odds ratio.

clinically important change to be compared, which can be readily converted into number-needed-to-treat (NNT) (e.g., the NNT for 1 patient to experience an improvement of ≥20 points). For example, in the DEFINE-HF (Dapagliflozin Effect on Symptoms and Biomarkers in Patients With Heart Failure) study, Nassif et al. (30) reported not only the mean difference between groups (3.7-point greater improvement in the KCCQ overall summary score with dapagliflozin versus placebo; $p = 0.037$) but also provided the distributions of changes. These analyses revealed that the proportion of patients who worsened was lower in the dapagliflozin arm (18% on dapagliflozin vs. 25% on placebo) yielding an NNT of 14 patients to be treated with dapagliflozin for one to not be clinically worse. Conversely, the difference in those who improved by ≥5 points was substantially higher in the dapagliflozin arm (43% vs. 33%), as were the proportions who have very large (≥20 points; 15% vs. 7%) clinical change, suggesting NNTs of ~10 for one patient to be either better, or a greater deal better, when treated with dapafligozin. Whereas the overall conclusion was unchanged, there is a statistically significant benefit in health status from the use of dapagliflozin compared with placebo; the health status benefits are more clearly communicated by presenting the distribution of response categories.

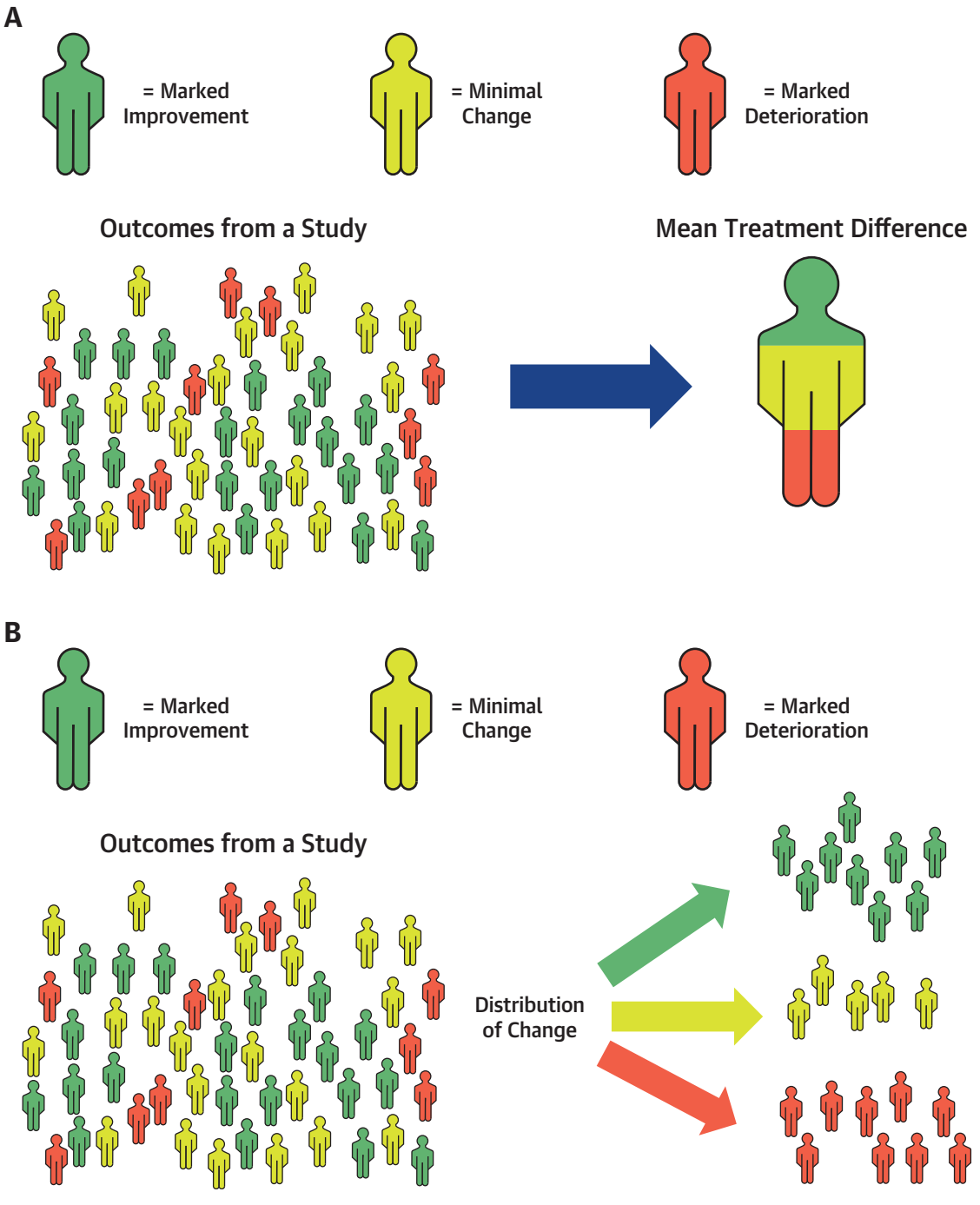
Occasionally, studies are designed without a baseline assessment from which changes in scores can be calculated. For such studies, the best way to present the results is by categories of clinical interpretability at a single point in time. For example, in the Tele-HF (Yale Heart Failure Telemonitoring Study) trial (31), the mean difference in KCCQ at 6 months between the tele-monitoring and control group was 2.5 points ($p = 0.02$). However, the proportions of patients who had fair to excellent (KCCQ overall summary scores of 50 to 100) versus very poor to fair (scores of 0 to 49), and good to excellent (scores of 75 to 100) versus very poor to good (scores of 0 to 74) health status demonstrated absolute differences at these thresholds of 6.9% and 3.7%, respectively. Thus, the number of patients needed to refer to tele-monitoring for 1 additional patient to have KCCQ scores of ≥50 or ≥75 was 15 and 27, respectively. Such an interpretation of these data provides a richer understanding of the value of referring patients to a tele-monitoring program.

A final challenge of using the KCCQ in clinical trials is that it can only be acquired on those who are alive at the time of the follow-up assessment. Excluding deaths from the analysis means that the results are only interpretable for those who would be alive at the follow-up time. If the mortality rate in the population of patients is both inconsequential and similar between treatments, then the comparisons between surviving patients will be unbiased. However, if there are a substantial number of deaths in an arm of a study, this will bias the estimation of follow-up health status in that arm upward—making it appear that the health status is better than it otherwise would have been had all patients survived. The reason for this is that patients with poor health status are more likely to die, and so when there are deaths, those with lower health status are systematically being removed from the analyses. Although more sophisticated analytic approaches exist to combine death and health status in joint models (32), Arnold et al. (33) have pioneered combining death and clinically important thresholds of KCCQ scores so that they can calculate the probability of being alive and well. Such a strategy directly assesses the most common question of patients concerning treatment—will I do better after treatment? All of these strategies—describing the distribution of patients who have clinically important changes, cross-sectional responder curves across clinical thresholds of outcomes, and combining health status and survival—are critical means for rendering the health status outcomes of trials more clinically interpretable.

USING THE KCCQ IN PATIENT CARE

At its core, the KCCQ is a standardized history focused on the impact of heart failure on patients' symptoms, function, and quality of life. Whereas clinicians can certainly elicit a better history, the practical challenges of busy clinics, fatigue, and distraction lead to variability over time within a given physician and, to an even greater extent, across

FIGURE 3 The Difference in Examining Average Group Changes Versus Proportions of Patients With Clinically Important Changes



The figure highlights that the population average **(A)** does not represent any individual patient in the trial. By describing the proportion of patients with different magnitudes of clinical change **(B)**, the clinical significance of the population average change is revealed.

physicians (15,34). In coronary disease, prior studies have shown substantial variability across physicians in their ability to consistently assess patients' symptom burden (34,35), and this variability can lead to marked differences in treatment (36). Accordingly, there is a compelling argument to be made for patient-reported health outcomes, such as the KCCQ, becoming a standard component of clinical visits.

This is particularly important in the era of rapidly accelerating tele-health and in group practices where care is distributed across different providers who may assign different NYHA functional classes to the same patient. In much the same way as clinicians review a patient's vital signs before a clinic visit, glancing at the KCCQ scores can rapidly inform the provider "how the patient is doing." Given the prognostic significance of KCCQ scores for death and hospitalization, it is important to know which scores (e.g., prior scores, current scores, or the change over time) are most prognostic. This was directly assessed by Pokharel et al. (12), who found that while previous scores and changes in scores were both associated with subsequent clinical events, the most recent score was most prognostic, suggesting that serial use of the KCCQ provides a continuously updating assessment of patients' risks.

A critical barrier to incorporating the KCCQ into clinical practice is clinicians' unfamiliarity with the interpretation of the instrument (37). Applying the methods described, however, enables clinicians to appreciate how patients are doing at a given point in time and how they have changed since their last assessment. The cross-sectional assessment is valuable in deciding whether further treatment intensification is needed to achieve a goal of maximizing KCCQ scores, whereas examining changes in scores is particularly useful for understanding the impact of instituting a new therapy. As an example, if a patient with HFrEF has a KCCQ overall summary score of 50 on an angiotensin-converting enzyme inhibitor and a beta-blocker and is euvolemic, then the physician might consider changing the angiotensin-converting enzyme inhibitor to sacubitril/valsartan, starting an sodium-glucose cotransporter-2 inhibitor or vericiguat, referring to cardiac rehabilitation, or recommending an intervention, such as cardiac resynchronization. The clinician can then monitor the impact of a new therapy on patients' health status at subsequent clinic visits with serial KCCQs. Improvements in scores of 5, 10, or 20 points provide a clearer quantification of the benefits of the therapeutic change and if no improvement is observed with one therapy, another can be tried.

More work is needed to define the duration of observation needed to assess whether a response to treatment has occurred. For example, whereas some interventions, such as diuretic therapy or percutaneous treatment of a dysfunctional valve may result in rapid improvement, other treatments that require left ventricular remodeling may take longer for their health status benefits to be achieved. In the DEFINE-HF study (30), for example, significant improvements in KCCQ scores were seen within 6 weeks, but even greater improvements were observed at 3 months. Moreover, failure to respond as expected after some interventions, such as cardiac resynchronization, may indicate a need for additional interventions, such as more optimal programming of the device. Future research defining the trajectory of health status improvement after different interventions will provide a better reference frame for defining when a patient is, or is not, responding to a given therapy.

Moreover, as clinicians begin to have a visceral understanding of what scores mean after correlating them with their patients' experiences, these scores can become an integral part of clinical care. A reasonable first step in evaluating KCCQ scores would be to focus on the overall summary score. If that score is low, then a deeper exploration into the domain scores can better explain whether it is the patients' symptoms, functional capacity, or quality of life that is most limiting their health status. Importantly, the actions a clinician might consider could vary if low symptom scores are driving the overall summary scale (e.g., more diuretics or better blood pressure control being needed) or if the symptoms and function are good, but the quality of life score is low, suggesting that a psychosocial intervention might be more important.

To achieve the clinical use of the KCCQ, however, requires developing the methods to routinely collect, score, and present the information within the clinical workflow. The collection and scoring of the KCCQ is most readily done through patient portals within the electronic health record, or by applications that can be completed on smart devices at home or in the waiting room. Once collected and scored, it is important to present the results to clinicians in a readily interpretable format. Whereas more research is needed, unpublished qualitative work at our institution suggests the most useful way to present the data may be plotting scores over time with labels on the y-axis (e.g., 0 to 24 = very poor to poor; 25 to 49 = poor to fair; 50 to 74 = fair to good; and 75 to 100 = good to excellent) to promote interpretation of the scores and color-coding of individual assessments (from red indicating very poor health status to green indicating excellent).

The final component of using the KCCQ in clinical care is to engage patients in completing the instrument and understanding its results. Researchers at the University of Utah, where the KCCQ was implemented into routine practice within the heart failure clinic, have recently reported qualitative research and survey results of patients' experiences in

completing the KCCQ as part of their clinic visits (38). Among the 24 participants interviewed, 23 (96%) felt that the use of the KCCQ made their evaluation more accurate, but only 15 (62%) felt that their providers effectively discussed their KCCQ results with them. In contrast to potential concerns about a Hawthorne effect, the qualitative analyses highlighted patients' perceptions that the use of the KCCQ enabled them to more effectively communicate with their providers about how their heart failure was impacting their lives. However, this study also revealed that the failure of providers to discuss their results with them undermined patients' perceived value of completing the questionnaires. Collectively, these data underscore the potential benefit to patients of routinely using the KCCQ in clinical care, but also the importance of clinicians understanding and sharing the results with their patients.

USING THE KCCQ IN POPULATION HEALTH

Once a health status instrument is integrated into routine clinical care, it is able to provide a key insight into the status of patients' health that is not otherwise available in the electronic health record and that is strongly associated with patients' risks for death and hospitalizations. It has previously been shown that there is marked variability across providers in the optimization of their patients' health status (39). A health system that has broad access to the health status of its patients with heart failure can readily create the means for the referral of patients doing poorly to more advanced specialists who might be able to identify new treatment opportunities to improve patients' health status and reduce their risks of clinical events.

CONCLUSIONS

The availability of psychometrically sound patient-reported outcomes measures, such as the KCCQ, offers an opportunity to more directly integrate the patient's experience—the impact of a disease on their symptoms, function, and quality of life—into all aspects of clinical care. The recent qualification of the KCCQ by the FDA (5,6) suggests that it will be increasingly used to test the benefits of new therapies. In fact, virtual trials without any in-person site visits, such as CHIEF-HF (A Study on Impact of Canagliflozin on Health Status, Quality of Life, and Functional Status in Heart Failure) (NCT04252287), are being launched to collect the KCCQ as the primary outcome of a clinical trial. However, the traditional presentation of health status results in clinical trials has precluded an intuitive assessment of the clinical importance of statistically significant results—a barrier that could be overcome by presenting the distribution of observed changes. Although the clinical use of formal health status assessments has been slow, there is growing interest in using such tools as measures of health care quality (40). Should physicians ultimately be held accountable for the health status of their patients, they will be faced with either collecting the data as part of an unfunded mandate or integrating the KCCQ into their clinical practice. Given the prognostic importance and the desire of patients to use the KCCQ in routine clinical care (38), engaging physicians to routinely use the KCCQ may not only improve the efficiency and quality of care, but may also enable participation in such quality assessment programs to be a simple by-product of routine practice, rather than an unfunded mandate and expense. Moreover, the routine use of the KCCQ can help support population health by readily identifying the most ill patients for whom more resources can be directed.

At the present time, the use of the KCCQ, beyond clinical trials, is more aspiration than reality. Although more research and experience are needed, particularly for the use of the KCCQ in clinical care and population health, the benefits of providing more efficient, more patient-centered care are great and justify the efforts to more seamlessly integrate the patient's "voice" into clinical practice.

AUTHOR RELATIONSHIP WITH INDUSTRY

Dr. Spertus owns the copyright to the Kansas City Cardiomyopathy Questionnaire. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

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